JEE-2007
Paper I

1. A resistance of 2 Ω is connected across one gap of a metre-bridge (the length of the wire is 100 cm) and an unknown resistance, greater than 2 Ω, is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any corrections, the unknown resistance is

(A) 3 Ω (B) 4 Ω (C) 5 Ω (D) 6 Ω

Answer ● ○ ○ ○

(A) (B) (C) (D)

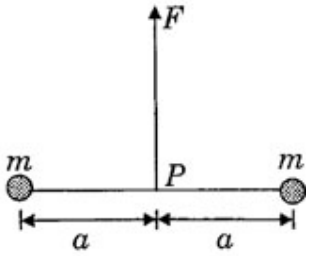
2. In an experiment to determine the focal length (f) of a concave mirror by the u-v method, a student places the object pin A on the principal axis at a distance x from the pole P. The student looks at the pin and its inverted image from a distance keeping his/her eye in line with PA. When the student shifts his/her eye towards left, the image appears to the right of the object pin. Then,

(A) $x < f$ (B) $f < x < 2f$ (C) $x = 2f$ (D) $x > 2f$

Answer ○ ● ○ ○

(A) (B) (C) (D)

3. Two particles of mass m each are tied at the ends of a light string of length $2a$. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance 'a' from the center P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surface. The magnitude of acceleration, when the separation between them becomes $2x$, is



(A) $\frac{F}{2m}\frac{a}{\sqrt{a^2-x^2}}$ (B) $\frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$

(C) $\frac{F}{2m}\frac{x}{a}$ (D) $\frac{F}{2m}\frac{\sqrt{a^2-x^2}}{x}$

Answer ○ ● ○ ○

(A) (B) (C) (D)

4. A long, hollow conducting cylinder is kept coaxially inside another long, hollow conducting cylinder of larger radius. Both the cylinders are initially electrically neutral.

(A) A potential difference appears between the two cylinders when a charge density is given to the inner cylinder

(B) A potential difference appears between the two cylinders when a charge density is given to the outer cylinder

(C) No potential difference appears between the two cylinders when a uniform line charge is kept along the axis of the cylinders

(D) No potential difference appears between the two cylinders when same charge density is given to both the cylinders

Answer ● ○ ○ ○

(A) (B) (C) (D)

5. Consider a neutral conducting sphere. A positive point charge is placed outside the sphere. The net charge on the sphere is then,

(A) negative and distributed uniformly over the surface of the sphere

(B) negative and appears only at the point on the sphere closest to the point charge

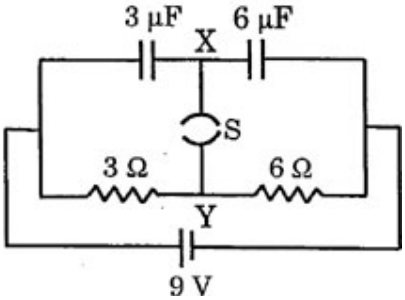
(C) negative and distributed non-uniformly over the entire surface of the sphere

(D) zero

Answer ○ ○ ○ ●

(A) (B) (C) (D)

6. A circuit is connected as shown in the figure with the switch S open. When the switch is closed, the total amount of charge that flows from Y to X is



(A) 0 (B) 54 μC

(C) 27 μC (D) 81 μC

Answer ○ ○ ● ○

(A) (B) (C) (D)

7. A ray of light traveling in water is incident on its surface open to air. The angle of incidence is θ, which is less than the critical angle. Then there will be

(A) only a reflected ray and no refracted ray

(B) only a refracted ray and no reflected ray

(C) a reflected ray and a refracted ray and the angle between them would be less than $180° - 2\theta$

(D) a reflected ray and a refracted ray and the angle between them would be greater than $180° - 2\theta$

Answer ○ (A) ○ (B) ● (C) ○ (D)

8. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is

(A) $E\left({}^{236}_{92}U\right) > E\left({}^{137}_{53}I\right) + E\left({}^{97}_{39}Y\right) + 2E(n)$

(B) $E\left({}^{236}_{92}U\right) < E\left({}^{137}_{53}I\right) + E\left({}^{97}_{39}Y\right) + 2E(n)$

(C) $E\left({}^{236}_{92}U\right) < E\left({}^{140}_{56}Ba\right) + E\left({}^{94}_{36}Kr\right) + 2E(n)$

(D) $E\left({}^{236}_{92}U\right) = E\left({}^{140}_{56}Ba\right) + E\left({}^{94}_{36}Kr\right) + 2E(n)$

Answer ● (A) ○ (B) ○ (C) ○ (D)

9. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm. The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is

(A) 802 nm (B) 823 nm (C) 1882 nm (D) 1648 nm

Answer ○ (A) ● (B) ○ (C) ○ (D)

10. STATEMENT-1

A block of mass m starts moving on a rough horizontal surface with a velocity v. It stops due to friction between the block and the surface after moving through a certain distance. The surface is now tilted to an angle of 30° with the horizontal and the same block is made to go up on the surface with the same initial velocity v. The decrease in the mechanical energy in the second situation is smaller than that in the first situation.

because

STATEMENT-2

The coefficient of friction between the block and the surface decreases with the increase in the angle of inclination.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ○ (B) ● (C) ○ (D)

11. STATEMENT-1

In an elastic collision between two bodies, the relative speed of the bodies after collision is equal to the relative speed before the collision.

because

STATEMENT-2

In an elastic collision, the linear momentum of the system is conserved.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ● (B) ○ (C) ○ (D)

OR

○ (A) ○ (B) ○ (C) ● (D)

12. STATEMENT-1

The formula connecting u, v and f for a spherical mirror is valid only for mirrors whose sizes are very small compared to their radii of curvature.

because

STATEMENT-2

Laws of reflection are strictly valid for plane surfaces, but not for large spherical surfaces.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ○ (B) ● (C) ○ (D)

13. STATEMENT-1

If the accelerating potential in an X-ray tube is increased, the wavelengths of the characteristic X-rays do not change.

because

STATEMENT-2

When an electron beam strikes the target in an X-ray tube, part of the kinetic energy is converted into X-ray energy.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ● (B) ○ (C) ○ (D)

14. The ratio x_1 / x_2 is

(A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Answer ○ ○ ● ○

(A) (B) (C) (D)

15. When disc B is brought in contact with disc A, they acquire a common angular velocity in time t. The average frictional torque on one disc by the other during this period is

(A) $\frac{2I\omega}{3t}$ (B) $\frac{9I\omega}{2t}$ (C) $\frac{9I\omega}{4t}$ (D) $\frac{3I\omega}{2t}$

Answer ● ○ ○ ○

(A) (B) (C) (D)

16. The loss of kinetic energy during the above process is

(A) $\frac{I\omega^2}{2}$ (B) $\frac{I\omega^2}{3}$ (C) $\frac{I\omega^2}{4}$ (D) $\frac{I\omega^2}{6}$

Answer ○ ● ○ ○

(A) (B) (C) (D)

17. The piston is now pulled out slowly and held at a distance $2L$ from the top. The pressure in the cylinder between its top and the piston will then be

(A) P_0 (B) $\frac{P_0}{2}$ (C) $\frac{P_0}{2} + \frac{Mg}{\pi R^2}$ (D) $\frac{P_0}{2} - \frac{Mg}{\pi R^2}$

Answer ● ○ ○ ○

(A) (B) (C) (D)

18. While the piston is at a distance $2L$ from the top, the hole at the top is sealed. The piston is then released, to a position where it can stay in equilibrium. In this condition, the distance of the piston from the top is

(A) $\left(\dfrac{2P_0\pi R^2}{\pi R^2 P_0 + Mg}\right)(2L)$

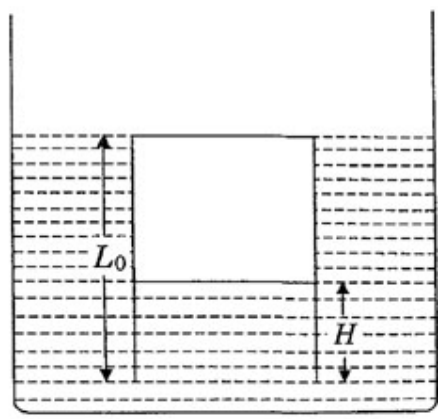
(B) $\left(\dfrac{P_0\pi R^2 - Mg}{\pi R^2 P_0}\right)(2L)$

(C) $\left(\dfrac{P_0\pi R^2 + Mg}{\pi R^2 P_0}\right)(2L)$

(D) $\left(\dfrac{P_0\pi R^2}{\pi R^2 P_0 - Mg}\right)(2L)$

Answer ○ (A) ○ (B) ○ (C) ● (D)

19. The piston is taken completely out of the cylinder. The hole at the top is sealed. A water tank is brought below the cylinder and put in a position so that the water surface in the tank is at the same level as the top of the cylinder as shown in the figure. The density of the water is ρ. In equilibrium, the height H of the water column in the cylinder satisfies



(A) $\rho g(L_0 - H)^2 + P_0(L_0 - H) + L_0P_0 = 0$

(B) $\rho g(L_0 - H)^2 - P_0(L_0 - H) - L_0P_0 = 0$

(C) $\rho g(L_0 - H)^2 + P_0(L_0 - H) - L_0P_0 = 0$

(D) $\rho g(L_0 - H)^2 - P_0(L_0 - H) + L_0P_0 = 0$

Answer ○ (A) ○ (B) ● (C) ○ (D)

20. Some physical quantities are given in **Column I** and some possible SI units in which these quantities may be expressed are given in **Column II**. Match the physical quantities in **Column I** with the units in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) GM_eM_s G – universal gravitational constant, M_e – mass of the earth, M_s – mass of the Sun	(p) (volt) (coulomb) (metre)
(B) $\frac{3RT}{M}$ R – universal gas constant, T – absolute temperature, M – molar mass	(q) (kilogram) (metre)^3 (second)$^{-2}$
(C) $\frac{F^2}{q^2B^2}$ F – force, q – charge, B – magnetic field	(r) (metre)^2 (second)$^{-2}$
(D) $\frac{GM_e}{R_e}$ G – universal gravitational constant, M_e – mass of the earth, R_e – radius of the earth	(s) (farad) (volt)^2 (kg)$^{-1}$

Answer

	p	q	r	s
A	●	●	○	○
B	○	○	●	●
C	○	○	●	●
D	○	○	●	●

21. **Column I** gives certain situations in which a straight metallic wire of resistance R is used and **Column II** gives some resulting effects. Match the statements in **Column I** with the statements in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) A charged capacitor is connected to the ends of the wire	(p) A constant current flows through the wire
(B) The wire is moved perpendicular to its length with a constant velocity in a uniform magnetic field perpendicular to the plane of motion	(q) Thermal energy is generated in the wire
(C) The wire is placed in a constant electric field that has a direction along the length of the wire	(r) A constant potential difference develops between the ends of the wire
(D) A battery of constant emf is connected to the ends of the wire	(s) Charges of constant magnitude appear at the ends of the wire

Answer

	p	q	r	s
A	○	●	○	○
B	○	○	●	●
C	○	○	○	●
D	●	●	●	○

22. Some laws / processes are given in **Column I**. Match these with the physical phenomena given in **Column II** and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	Column II
(A) Transition between two atomic energy levels	(p) Characteristic X-rays
(B) Electron emission from a material	(q) Photoelectric effect
(C) Mosley's law	(r) Hydrogen spectrum
(D) Change of photon energy into kinetic energy of electrons	(s) β- decay

Answer

	p	q	r	s
A	●	○	●	○
B	○	●	○	●
C	●	○	○	○
D	○	●	○	○

23. The species having bond order different from that in CO is

(A) NO^- (B) NO^+

(C) CN^- (D) N_2

Answer ● ○ ○ ○

(A) (B) (C) (D)

24. Among the following, the paramagnetic compound is

(A) Na_2O_2 (B) O_3

(C) N_2O (D) KO_2

Answer ○ ○ ○ ●

(A) (B) (C) (D)

25. Extraction of zinc from zinc blende is achieved by

(A) electrolytic reduction

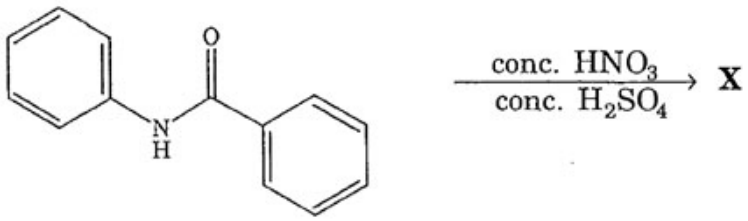
(B) roasting followed by reduction with carbon

(C) roasting followed by reduction with another metal

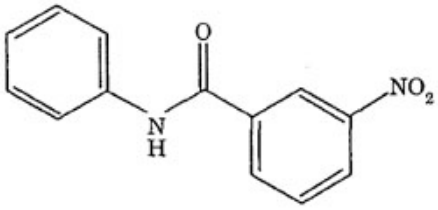
(D) roasting followed by self-reduction

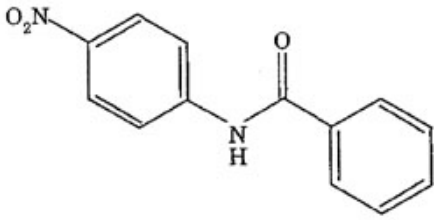
Answer ○ (A) ● (B) ○ (C) ○ (D)

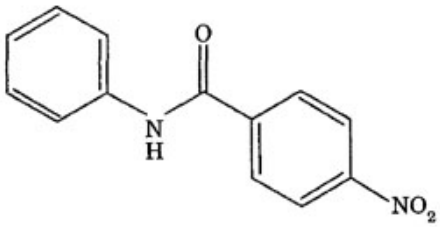
26. In the following reaction,

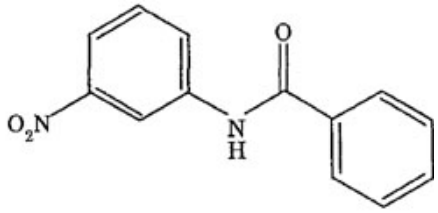


the structure of the major product '**X**' is

(A)


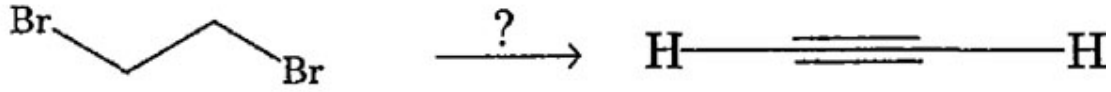
(B)


(C)


(D)


Answer ○ (A) ● (B) ○ (C) ○ (D)

27. The reagent(s) for the following conversion,



is/are

(A) alcoholic KOH

(B) alcoholic KOH followed by $NaNH_2$

(C) aqueous KOH followed by $NaNH_2$

(D) Zn/CH_3OH

Answer ○ ● ○ ○
(A) (B) (C) (D)

28. The number of structural isomers for C_6H_{14} is

(A) 3 (B) 4

(C) 5 (D) 6

Answer ○ ○ ● ○
(A) (B) (C) (D)

29. The percentage of p-character in the orbitals forming P–P bonds in P_4 is

(A) 25 (B) 33

(C) 50 (D) 75

Answer ○ ○ ○ ●

(A) (B) (C) (D)

30. When 20 g of naphthoic acid ($C_{11}H_8O_2$) is dissolved in 50 g of benzene ($K_f = 1.72$ K kg mol^{-1}), a freezing point depression of 2 K is observed. The van't Hoff factor (i) is

(A) 0.5 (B) 1

(C) 2 (D) 3

Answer ● ○ ○ ○

(A) (B) (C) (D)

31. The value of $\log_{10}K$ for a reaction $A \rightleftharpoons B$ is

(Given : $\Delta_r H^\circ_{298K} = -54.07$ kJ mol^{-1}, $\Delta_r S^\circ_{298K} = 10$ $JK^{-1}mol^{-1}$ and $R = 8.314$ $JK^{-1}mol^{-1}$; $2.303 \times 8.314 \times 298 = 5705$)

(A) 5 (B) 10

(C) 95 (D) 100

Answer ○ ● ○ ○

(A) (B) (C) (D)

32. STATEMENT-1 : Boron always forms covalent bond.

because

STATEMENT-2 : The small size of B^{3+} favours formation of covalent bond.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ● ○ ○ ○

(A) (B) (C) (D)

33. STATEMENT-1 : In water, orthoboric acid behaves as a weak monobasic acid.

because

STATEMENT-2 : In water, orthoboric acid acts as a proton donor.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

34. STATEMENT-1 : *p*–Hydroxybenzoic acid has a lower boiling point than *o*-hydroxybenzoic acid.

because

STATEMENT-2 : *o*–Hydroxybenzoic acid has intramolecular hydrogen bonding.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ○ (B) ○ (C) ● (D)

35. STATEMENT-1 : Micelles are formed by surfactant molecules above the critical micellar concentration (CMC).

because

STATEMENT-2 : The conductivity of a solution having surfactant molecules decreases sharply at the CMC.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ● (B) ○ (C) ○ (D)

36. Argon is used in arc welding because of its

(A) low reactivity with metal

(B) ability to lower the melting point of metal

(C) flammability

(D) high calorific value

Answer ● ○ ○ ○

(A) (B) (C) (D)

37. The structure of XeO_3 is

(A) linear (B) planar (C) pyramidal (D) T-shaped

Answer ○ ○ ● ○

(A) (B) (C) (D)

38. XeF_4 and XeF_6 are expected to be

(A) oxidizing (B) reducing (C) unreactive (D) strongly basic

Answer ● ○ ○ ○

(A) (B) (C) (D)

OR

○ ● ○ ○

(A) (B) (C) (D)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

39. The total number of moles of chlorine gas evolved is

(A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0

Answer ○ (A) ● (B) ○ (C) ○ (D)

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A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

40. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is

(A) 200 (B) 225 (C) 400 (D) 446

Answer ○ (A) ○ (B) ○ (C) ● (D)

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A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

41. The total charge (coulombs) required for complete electrolysis is

(A) 24125 (B) 48250 (C) 96500 (D) 193000

Answer ○ (A) ○ (B) ○ (C) ● (D)

42. Match the complexes in **Column I** with their properties listed in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(p) geometrical isomers
(B) $[Pt(NH_3)_2Cl_2]$	(q) paramagnetic
(C) $[Co(H_2O)_5Cl]Cl$	(r) diamagnetic
(D) $[Ni(H_2O)_6]Cl_2$	(s) metal ion with +2 oxidation state

Answer

	p	q	r	s
A	●	●	○	●
B	●	○	●	●
C	○	●	○	●
D	○	●	○	●

43. Match the chemical substances in **Column I** with type of polymers/type of bonds in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) cellulose	(p) natural polymer
(B) nylon-6, 6	(q) synthetic polymer
(C) protein	(r) amide linkage
(D) sucrose	(s) glycoside linkage

Answer

	p	q	r	s
A	●	○	○	●
B	○	●	●	○
C	●	○	●	○
D	○	○	○	●

44. Match gases under specified conditions listed in **Column I** with their properties/laws in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) hydrogen gas ($P = 200$ atm, $T = 273$ K)	(p) compressibility factor $\neq 1$
(B) hydrogen gas ($P \sim 0$, $T = 273$ K)	(q) attractive forces are dominant
(C) CO_2 ($P = 1$ atm, $T = 273$ K)	(r) $PV = nRT$
(D) real gas with very large molar volume	(s) $P(V-nb) = nRT$

Answer

	p	q	r	s
A	●	○	○	●
B	○	○	●	○
C	●	●	○	○
D	●	○	○	●

45. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}$, 2β be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

(A) $\frac{2}{9}(p-q)(2q-p)$ (B) $\frac{2}{9}(q-p)(2p-q)$

(C) $\frac{2}{9}(q-2p)(2q-p)$ (D) $\frac{2}{9}(2p-q)(2q-p)$

Answer ○ ○ ○ ●

(A) (B) (C) (D)

46. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and

$$\lim_{t \to x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$$

for each $x > 0$. Then $f(x)$ is

(A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$ (C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$

Answer ● ○ ○ ○

(A) (B) (C) (D)

47. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{1}{5}$

Answer ○ ○ ● ○

(A) (B) (C) (D)

48. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$

(A) on the left of $x = c$ (B) on the right of $x = c$

(C) at no point (D) at all points

Answer ● ○ ○ ○

(A) (B) (C) (D)

49. $\lim_{x \to \frac{\pi}{4}} \dfrac{\int_{2}^{\sec^2 x} f(t)\, dt}{x^2 - \frac{\pi^2}{16}}$ equals

(A) $\frac{8}{\pi} f(2)$ (B) $\frac{2}{\pi} f(2)$ (C) $\frac{2}{\pi} f\left(\frac{1}{2}\right)$ (D) $4 f(2)$

Answer ● ○ ○ ○

(A) (B) (C) (D)

50. A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is

(A) $x^2 \operatorname{cosec}^2\theta - y^2 \sec^2\theta = 1$ (B) $x^2 \sec^2\theta - y^2 \operatorname{cosec}^2\theta = 1$

(C) $x^2 \sin^2\theta - y^2 \cos^2\theta = 1$ (D) $x^2 \cos^2\theta - y^2 \sin^2\theta = 1$

Answer ● ○ ○ ○

(A) (B) (C) (D)

51. The number of distinct real values of λ, for which the vectors $-\lambda^2\hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2\hat{k}$ are coplanar, is

(A) zero (B) one

(C) two (D) three

Answer ○ ○ ● ○

(A) (B) (C) (D)

52. A man walks a distance of 3 units from the origin towards the north-east ($N\ 45^\circ\ E$) direction. From there, he walks a distance of 4 units towards the north-west ($N\ 45^\circ\ W$) direction to reach a point P. Then the position of P in the Argand plane is

(A) $3e^{i\pi/4} + 4i$

(B) $(3 - 4i)e^{i\pi/4}$

(C) $(4 + 3i)e^{i\pi/4}$

(D) $(3 + 4i)e^{i\pi/4}$

Answer ○ (A) ○ (B) ○ (C) ● (D)

53. The number of solutions of the pair of equations

$$2\sin^2\theta - \cos 2\theta = 0$$
$$2\cos^2\theta - 3\sin\theta = 0$$

in the interval $[0, 2\pi]$ is

(A) zero

(B) one

(C) two

(D) four

Answer ○ (A) ○ (B) ● (C) ○ (D)

54. Let $H_1, H_2, \cdots, H_n$ be mutually exclusive and exhaustive events with $P(H_i) > 0$, $i = 1, 2, \ldots, n$. Let E be any other event with $0 < P(E) < 1$.

STATEMENT-1 : $P(H_i \mid E) > P(E \mid H_i) \cdot P(H_i)$ for $i = 1, 2, \ldots, n$.

because

STATEMENT-2 : $\sum_{i=1}^{n} P(H_i) = 1$.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ○ (B) ○ (C) ● (D)

55. Tangents are drawn from the point $(17, 7)$ to the circle $x^2 + y^2 = 169$.

STATEMENT-1 : The tangents are mutually perpendicular.

because

STATEMENT-2 : The locus of the points from which mutually perpendicular tangents can be drawn to the given circle is $x^2 + y^2 = 338$.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ● ○ ○ ○

(A) (B) (C) (D)

56. Let the vectors $\vec{PQ}$, $\vec{QR}$, $\vec{RS}$, $\vec{ST}$, $\vec{TU}$ and $\vec{UP}$ represent the sides of a regular hexagon.

STATEMENT-1 : $\vec{PQ} \times \left(\vec{RS} + \vec{ST}\right) \neq \vec{0}$.

because

STATEMENT-2 : $\vec{PQ} \times \vec{RS} = \vec{0}$ and $\vec{PQ} \times \vec{ST} \neq \vec{0}$.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

57. Let $F(x)$ be an indefinite integral of $\sin^2 x$.

STATEMENT-1 : The function $F(x)$ satisfies $F(x+\pi) = F(x)$ for all real x.

because

STATEMENT-2 : $\sin^2(x+\pi) = \sin^2 x$ for all real x.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ (A) ○ (B) ○ (C) ● (D)

M58-60 : Paragraph for Question Nos. 58 to 60

Let V_r denote the sum of the first r terms of an arithmetic progression (A.P.) whose first term is r and the common difference is $(2r-1)$. Let

$$T_r = V_{r+1} - V_r - 2 \text{ and } Q_r = T_{r+1} - T_r \text{ for } r = 1, 2, \ldots$$

58. The sum $V_1 + V_2 + \ldots + V_n$ is

(A) $\frac{1}{12}n(n+1)(3n^2 - n + 1)$

(B) $\frac{1}{12}n(n+1)(3n^2 + n + 2)$

(C) $\frac{1}{2}n(2n^2 - n + 1)$

(D) $\frac{1}{3}(2n^3 - 2n + 3)$

Answer ○ (A) ● (B) ○ (C) ○ (D)

59. T_r is always

(A) an odd number
(B) an even number
(C) a prime number
(D) a composite number

Answer ○ ○ ○ ●
(A) (B) (C) (D)

60. Which one of the following is a correct statement?

(A) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 5
(B) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 6
(C) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 11
(D) $Q_1 = Q_2 = Q_3 = \ldots$

M61-63 : Paragraph for Question Nos. 61 to 63

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

Answer ○ ● ○ ○
(A) (B) (C) (D)

61. The ratio of the areas of the triangles PQS and PQR is

(A) $1 : \sqrt{2}$
(B) $1 : 2$
(C) $1 : 4$
(D) $1 : 8$

Answer ○ ○ ● ○
(A) (B) (C) (D)

62. The radius of the circumcircle of the triangle PRS is

(A) 5
(B) $3\sqrt{3}$
(C) $3\sqrt{2}$
(D) $2\sqrt{3}$

Answer ○ ● ○ ○
(A) (B) (C) (D)

63. The radius of the incircle of the triangle PQR is

(A) 4 (B) 3 (C) $\frac{8}{3}$ (D) 2

Answer

(A)	(B)	(C)	(D)
○	○	○	●

64. Consider the following linear equations

$$ax + by + cz = 0$$
$$bx + cy + az = 0$$
$$cx + ay + bz = 0$$

Match the conditions/expressions in **Column I** with statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(p) the equations represent planes meeting only at a single point.
(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(q) the equations represent the line $x = y = z$.
(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$	(r) the equations represent identical planes.
(D) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$	(s) the equations represent the whole of the three dimensional space.

Answer

	p	q	r	s
A	○	○	●	○
B	○	●	○	○
C	●	○	○	○
D	○	○	○	●

65. In the following $[x]$ denotes the greatest integer less than or equal to x.

Match the functions in **Column I** with the properties in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) $x\|x\|$	(p) continuous in $(-1, 1)$
(B) $\sqrt{\|x\|}$	(q) differentiable in $(-1, 1)$
(C) $x+[x]$	(r) strictly increasing in $(-1, 1)$
(D) $\|x-1\|+\|x+1\|$	(s) not differentiable at least at one point in $(-1, 1)$

Answer

	p	q	r	s
A	●	●	●	○
B	●	○	○	●
C	○	○	●	●
D	●	●	○	○

66. Match the integrals in **Column I** with the values in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) $\int_{-1}^{1} \frac{dx}{1+x^2}$	(p) $\frac{1}{2}\log\left(\frac{2}{3}\right)$
(B) $\int_{0}^{1} \frac{dx}{\sqrt{1-x^2}}$	(q) $2\log\left(\frac{2}{3}\right)$
(C) $\int_{2}^{3} \frac{dx}{1-x^2}$	(r) $\frac{\pi}{3}$
(D) $\int_{1}^{2} \frac{dx}{x\sqrt{x^2-1}}$	(s) $\frac{\pi}{2}$

Answer

	p	q	r	s
A	○	○	○	●
B	○	○	○	●
C	●	○	○	○
D	○	○	●	○